

SHASHANK RAJESH KULKARNI

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EDUCATION

Rochester Institute of Technology, Rochester, NY - MS, Cybersecurity (GPA - 3.93/4)

Dec 2025

Gandhi Institute of Technology and Management, India - B.Tech, Computer Science

June 2022

PROJECTS

Cybersecurity Home Lab

- Designed and maintained a comprehensive **cybersecurity home lab** with multiple virtual machines - Active Directory, Splunk (SIEM), Nessus (Vulnerability management), Suricata (IDS/IPS), Sysmon, TheHive (incident response), pfSense (firewall), Kali Linux (attack machine), and various intentionally vulnerable systems, **automated** using **Ansible** to streamline configuration and deployment.

Consulting Services for The Charity Project of Henrietta

- Conducted a comprehensive IT security assessment for a multi-location non-profit, **identifying critical risks** and developing a 3-year **NIST**-compliant cybersecurity roadmap, Performed Business Impact Analysis (BIA) and root cause analysis to prioritize **risk remediation**, defined RTOs/RPOs, and improved disaster recovery, cloud migration, and data protection measures.

Analysis of Lightweight cryptography Encryption and File Transfer in AWS Cloud

- Developed and deployed a lightweight **encryption tool** using AES-GCM, ChaCha20-Poly1305, and AES-CTR with SHA-256 for secure file transfer in a resource-constrained **cloud environment**, achieving optimal encryption performance for files up to 1GB.

CERTIFICATIONS

- Certified Ethical Hacker | EC-Council
- CompTIA Security+
- Amazon Web Services (AWS) Certified Solutions Architect Associate
- CompTIA Cybersecurity Analyst (CySA+)

Ongoing

Ongoing

TECHNICAL SKILLS

- Skills:** SOC Operations, Threat Hunting, Risk Management, Incident Response, Vulnerability Assessment, Security Policy Development, Security Compliance (NIST, HIPAA, GDPR, PCI-DSS, COPPA, FERPA, FTC)
- Programming and Automation:** Python, SQL, PowerShell, Bash, Ansible
- Tools:** SIEM (Graylog, Splunk, Wazuh, Security Onion), IDS/IPS (Snort, Suricata), SOAR (Shuffle), Firewall (PfSense), Threat Intelligence (MISP, VirusTotal)
- Operating Systems:** Linux (Ubuntu, Kali), Windows (Server & Client)
- Network and Web Security:** TCP/IP, Nmap, Recon-ng, Nessus, Wireshark, Cisco Packet Tracer, Burp Suite, Postman, Active Directory Security, MITRE ATT&CK Framework
- Cloud Technologies:** AWS (IAM, Security Hub, GuardDuty, CloudTrail, KMS, Security Group), Kubernetes

EXPERIENCE

Tutor | Rochester Institute of Technology | Rochester, NY

Aug 2024 - Present

- Instructed **50+ students** on penetration testing methodologies, covering threat modeling, risk assessment, and exploitation using tools like Metasploit, Nmap, and Burp Suite.
- Explained cryptographic algorithms (AES, RSA, DES) and helped students understand encryption protocols and key management.

Implementation Engineer | ENEA AdaptiveMobile Security | Remote

June 2022 - April 2023

- Migrated legacy infrastructure to AWS, reducing operational costs by **15%** and improving security by enhancing IAM policies, Security Groups, and encryption mechanisms.
- Managed and scaled infrastructure supporting **2.2 billion** global subscribers. **Automated** SMS traffic analysis using shell scripts, reducing manual workload by **20%** and improving log analysis speed.

Cybersecurity Trainee Intern | Castellum Labs | Remote

May 2022

- Conducted OSINT investigations and passive footprinting, identifying **20+ security vulnerabilities** in client environments and providing actionable recommendations.
- Assisted in risk assessments and security control implementation, contributing to a 30% reduction in security incidents across client networks.

Software Development Engineer Intern | Prakya | Remote

July 2021 - April 2022

- Developed automated tests for a web application, achieving **70%** test coverage on core functionalities while integrating **secure coding** practices to identify and mitigate vulnerabilities during testing and **mentoring incoming interns**